Evaluating data extraction error by a large language model from randomised controlled trials: a large-scale empirical study

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Abstract

Objective To examine the potential errors of a general large language model (LLM) (ie, Claude 3.5 Sonnet) on data extraction from randomised controlled trials (RCTs).

Design and setting An empirical study comparing Claude 3.5 Sonnet extractions against a human-performed verification dataset. The extraction tasks for Claude 3.5 Sonnet were based solely on original RCT portable document format (PDF) files. For PDFs that could not be directly extracted by Claude 3.5 Sonnet, optical character recognition was employed to convert them into text format before extraction.

Participants A random sample of 664 trials was selected from a well-established trial bank and a final data pool was established based on rigorous manual cross-checking as a reference standard.

Data sources PubMed, EMBASE, Scopus, Web of Science (all databases) and the Cochrane Central Register of Controlled Trials (CENTRAL) up to February 2023.

Eligibility criteria for selecting studies RCTs on children involving medication and adverse events.

Main outcome measures Claude 3.5 Sonnet was applied to extract the basic information (eg, trial design, population information and source of funding) and adverse outcomes (ie, name of adverse events, number of events). Claude 3.5 Sonnet outputs were compared against the final data pool and all errors were recorded. Results are presented as error rates and with 95% CI, estimated using a generalised linear mixed model.

Results For the 664 trials, a total of 23 069 data cells were extracted via Claude 3.5 Sonnet, with 10 624 for basic information and 12 445 for adverse outcomes. The overall error rate for data extraction was 6.6% (95% CI 5.4% to 8.2%), with 5.7% (95% CI 5.2% to 6.1%) in basic information and 7.6% (95% CI 4.9% to 11.8%) in adverse outcomes. When stratified the 1542 total errors by error types, misallocation (assigning data to incorrect fields; 57.1%, 881/1542) and missed or omitted data (incomplete extraction of available data; 23.2%, 357/1542) accounted for the two most frequent errors, with misallocation occurring more in basic information (53.3%, 470/881), while missed or omitted data occurred more in adverse outcomes (96.1%, 343/357). Post hoc analysis examining the association between trial reporting quality (assessed using Consolidated Standards of Reporting Trials (CONSORT) 2025 and LLM data extraction error rates indicated that higher CONSORT adherence was associated with lower extraction error rates.

Conclusions The data extraction error of Claude was relatively low, but it alerts LLM applications in evidence synthesis. Detailed checking for LLM outputs should be the primary consideration for evidence synthesisers.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Evidence synthesis relies heavily on accurate data extraction from randomised controlled trials, but manual extraction is resource-intensive and error-prone. Large language models (LLMs) have shown promise for improving extraction efficiency, but their accuracy, error types, and limitations in medical contexts remain poorly characterised.

WHAT THIS STUDY ADDS

⇒ In this study of data extraction from 664 randomised controlled trials using Claude 3.5 Sonnet, the overall error rate was 6.6%. These errors could be grouped into five categories based on their mechanisms: false absence reporting, misallocation, fabricated data, missed or omitted data and numerical error. Additionally, Claude 3.5 Sonnet showed inconsistent error patterns across different data element types and, in post hoc exploratory analyses, exhibited a higher propensity for extraction errors in trials with suboptimal reporting quality.

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HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ These findings suggest that Claude could serve as a valuable data extraction tool in evidence synthesis, potentially reducing workload and costs for research teams while enabling more timely and accurate evidence synthesis for clinical decision-making. However, extraction errors by LLMs and their underlying mechanisms still warrant careful consideration, and future research should investigate how these LLM-based errors affect evidence synthesis, such as meta-analyses.

Introduction

In evidence-based medicine, systematic reviews provide a rigorous approach to synthesising evidence from clinical trials and are widely used to improve the estimate of medication-related harm estimates, addressing limited sample sizes and low event rates in individual trials. Notably, the validity of evidence syntheses is closely linked to accurate data extraction.[1] Current guidelines recommend double extraction as the best practice to minimise human error.[2] Even so, data extraction errors remain common because this process remains resource-intensive and susceptible to human error. A trial showed that for binary medication-related adverse outcome extraction, double extraction reduced error rates by 19.8% compared with single extraction, yet the error rate at the study level remained 44.9%.[3] More broadly, a reproducibility study showed that the data extraction error rate was 17.0% at the study level and 66.8% at the meta-analysis level.[1] This error may impact the direction and significance of results, undermine the reliability of evidence syntheses, and potentially compromise clinical decision-making.

To address these challenges, publicly accessible large language models (LLMs), such as Claude and ChatGPT, that can interact with external files (eg, portable document format (PDF) documents of published studies), offer transformative opportunities. These LLMs have remarkable capabilities in capturing complex contextual relationships and processing and analysing vast text corpora.[4] Previous evaluations reported accuracy rates of LLMs for data extraction ranging from 72% to 100% at the cell level,[5–7] compared with human extraction data. However, these studies have typically used small samples and relied on existing systematic review datasets as reference standards, which limits the generalisability of their findings to real-world practice, particularly given the stochastic nature of LLM outputs.[8] Notably, Gartlehner *et al* found that LLM-assisted extraction with human verification achieved similar accuracy (difference within 2%) to manual extraction with verification, but with greater efficiency.[9] These findings collectively demonstrate the promising prospects of applying LLMs for data extraction.

Harnessing LLMs for data extraction is a key initiative that streamlines evidence synthesis, but it also brings new drawbacks. Recent studies have suggested that LLMs produce unique categories of data extraction errors not typically observed in human extraction, such as 'hallucination' error (ie, generating fabricated information),[5 9] which might compromise the reliability of evidence-based medicine findings in evidence synthesis, potentially misleading clinical decision-making. Given these concerns, comprehensive evaluation of the frequency and nature of these errors across different data elements is essential for establishing when and how LLM-assisted extraction can be reliably integrated into systematic review workflows. Cell-level assessment provides the greater granularity for this evaluation: it examines each extracted data point individually, maximising sensitivity to detect error patterns and is closer to the actual workflow of extracting tables/fields. While cell-level errors do not directly equate to meta-analysis bias, they may affect effect estimates through error classification (eg, misallocation, missed extraction).

In this context, we conducted a large-scale empirical study of the error rates and types of data extraction by Claude 3.5 Sonnet, one of the most advanced LLMs, to extract basic information and adverse outcomes from randomised controlled trials (RCTs). We propose an empirical classification of the errors made by Claude 3.5 Sonnet during data extraction to help systematic review authors better understand the potential error mechanisms of LLMs in the extraction process, which may enable targeted improvements in subsequent tasks.

Methods

This empirical study is part of a larger research programme, the PROMISE (Promoting Medication Safety) project, an ongoing initiative dedicated to collecting RCTs investigating medication-related harms in children.[10] The study followed the Standards for Reporting of Diagnostic Accuracy Studies with artificial intelligence (AI) guidelines.[11]

Data source and gold standard

Our work was conducted based on a well-established trial bank developed from RCTs focused on adverse events retrieved from PubMed, Embase, Scopus, Web of Science (all databases), and the Cochrane Central Register of Controlled Trials (CENTRAL) up to February 2023. This Trial Bank included 13 650 eligible RCTs in which at least one arm involved a pharmaceutical compound or biological agent (the screening process is detailed in the online supplemental Figure S1). Data extraction for these trials employed a hybrid AI-human approach: Claude 2 Sonnet was used for the initial data extraction, followed by iterative human verification by well-trained student volunteers, those who achieved ≥90% agreement during training. The Trial Bank comprised two modules: basic information and adverse events. Prior to the current study, the basic information underwent two rounds of cross-checking by well-trained volunteers, achieving an estimated agreement of 96.1% after the second round. In contrast, the adverse events were extracted but not yet verified. Further details are available in our previous publication.[10]

To establish the final data pool (gold standard) for validating the accuracy of LLM data extraction, we implemented the following steps: in step one, we selected 5% of trials (n=682) from the Trial Bank using simple random sampling. In step two, the basic information was checked against the original publications by our more experienced team members (HC, YQY, JZL, YXW, YFZ), with 96.7% agreement. In step three, the adverse events underwent two rounds of cross-checking against the original publications by three authors (SF, MC, ZY), with 97.4% final agreement. Finally, after excluding 18 trials with abstracts only, the remaining 664 RCTs with repeatedly checked and corrected data constituted our final data pool, as the gold standard. The complete dataset used in this study is available at the Open Science Framework.[12]

Selection of LLM and design of prompt engineering

At the time of the study initiation, Claude 3.5 Sonnet and ChatGPT-4 were the most advanced LLMs. These artificial intelligence tools did not require complex coding knowledge and could be used easily by laypeople. We conducted a pilot study to refine

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prompts and select the optimal LLM. This was done in four steps. We first developed two primary prompt types for extraction tasks[4]: (a) the prompt designed to extract a single data element from RCT articles and (b) the prompt designed to extract all data elements from RCT articles simultaneously. We further conducted iterative testing by Claude, progressively refining the prompts for each task. Then, based on the refined prompts and 10 randomly selected RCTs,[13–22] we conducted data extraction performance tests with Claude 3.5 Sonnet and ChatGPT-4. Finally, Claude 3.5 Sonnet was chosen as the data extraction tool due to its superior testing results in our pilot study (see online supplemental table 1). For the extraction process, we input these refined prompts alongside PDF documents into Claude. To minimise memory retention bias, each prompt was executed in a new session. The final prompts for each data element are presented in online supplemental Boxes 1–15.

LLM-based data extraction

Our goal was to extract two sets of information from each RCT via the preselected LLM: (1) basic information: title, year of publication, the name of published journal, disease, region of the trial, number of centres of the trial, source of funding, registration status, registration number, design of the study, the name of intervention, the name of control, sample size, the mean/median of population age, the range of population age and data sharing statements and (2) adverse events (names and numbers of adverse events in intervention and comparison groups). Adverse events were defined as 'any untoward medical occurrence in a patient or subject in clinical practice' which consists of adverse events, adverse effects, adverse reactions, harms and complications in pharmaceutical interventions.[23] If trials reported no adverse events, we recorded double-zero events (0 events in treatment vs 0 events in control). All 17 data elements and their detailed definitions are presented in online supplemental table 2.

For each RCT, we checked the recognisability of the PDF files and used optical character recognition software (ABBYY FineReader PDF 15) to convert non-recognisable PDF files (eg, picture) into identifiable text. All extracted information was based on content explicitly reported or reasonably derivable from the original PDF; we did not extract data from supplementary materials, trial registries or clinical study reports. The data extraction tasks of Claude 3.5 Sonnet were performed by 11 well-trained undergraduate or postgraduate student team members (XYZ, XWQ, HFL, SCJ, ZY, YQL, ZYP, MC, WQZ, XHL and ZYC). Three authors (SF, ZY, MC) subsequently reviewed the extracted data by Claude 3.5 Sonnet with the pre-established gold standard, followed by a detailed checking of discrepancies between Claude 3.5 Sonnet outputs and the gold standard, and calculated all extraction errors. As these errors could have different mechanisms, we classified them into different types on the basis of consensus after summarising the error information (online supplemental figure S2).

Definition of error types

We compared the initial data extracted by Claude with the verified gold standard, marking inconsistencies as extraction errors. We have systematically categorised the errors made by the LLM during data extraction tasks into five distinct types: false absence reporting, misallocation, fabricated data, missed or omitted data and numerical error, based on frameworks for human data extraction errors established by our group and for LLM-assisted extraction errors recently proposed by Gartlehner *et al*.[1 8] Table 1

Table 1 Descriptions of the different types of errors for data extraction with Claude 3.5 Sonnet

Types of error	Description	Example
False absence reporting	The LLM failed to recognise data that were actually present in the study report and erroneously extracted them as 'not reported' or 'not available'.	In RCT[25] that examined propofol use in paediatric anaesthesia, the authors explicitly stated 'Source of funding: None' at the end of the main text. However, during extraction, the LLM incorrectly reported 'funding information not reported in the PDF, despite this explicit statement being present'.
Misallocation	The LLM incorrectly filled a target data field with content extracted from elsewhere in the text, regardless of whether the correct information for that field was present in the study.	In RCT[26] that investigated whether IFN-α treatment induces clinical or serological manifestations of autoimmune liver disease, the source PDF did not explicitly state the trial conduct location. During data extraction, the LLM incorrectly identified the authors' affiliations (such as 'Naples, Italy; Padova, Italy; London, England') as the study conduct locations, conflating author institutional addresses with trial sites.
Fabricated data	Data that were not available in the study report were erroneously generated (hallucinated) by the LLM.	In RCT[27] on beclomethasone dipropionate and formoterol pharmacokinetics in children with asthma, the PDF contained no mention of '2013' anywhere in the document. However, when extracting the publication year, the LLM incorrectly generated 2013 as the year of publication.
Missed or omitted data	Data that were provided in the study report, some portions of which were exactly extracted by the LLM, while others were missed or omitted.	In RCT[28] that compared postoperative analgesic effects of levobupivacaine, lidocaine and saline, the study design clearly described three intervention groups. However, during intervention extraction, the LLM identified only levobupivacaine as the intervention drug, completely omitting lidocaine, thus providing an incomplete list of interventions despite all being clearly stated in the methods section.
Numerical error	Failure to accurately extract numerical values despite recognising their presence, potentially due to recognition errors, format parsing issues, unit conversion mistakes, calculation errors, or other processing failures.	In the RCT[29] investigated gamma-globulin treatment for recurrent acute otitis media, mean ages were reported as 15.4 months for the treatment group (n=22) and 14.1 months for the control group (n=22). According to our calculation formula, the correct pooled mean age should be 14.75 months ((15.4×22+14.1×22)/(22+22)). However, Claude 3.5 Sonnet incorrectly computed this as 13.75 months.

LLM, large language model; RCT, randomised controlled trial.

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provides the definitions and detailed examples of these five types of data extraction errors.[24–28]

Outcomes

The primary outcomes were the extraction errors and error rates by Claude 3.5 Sonnet for overall data, basic information, adverse outcomes and each data element. Secondary outcomes included the proportion of each error type and its distribution across these different data extraction elements.

Sample size calculation

In a previous proof-of-concept study, Claude 2 achieved a minimum error rate of 3.1% for repeated data extraction.[5] In the current study, we aimed to estimate the overall data extraction error rate as well as the rate for each individual data element, assuming Claude 3.5 Sonnet would have the same extraction error rate across all data elements. To obtain a two-sided 95% CI with a width of approximately 3 percentage points, given an observed error rate of 3.1%, we determined that at least 535 data units per data element would be required. The sample size calculation was derived from the following formula[29]:

$$2 \cdot z_{1-\alpha/2} \cdot \sqrt{\frac{p(1-p)}{n} + \frac{z^2_{1-\alpha/2}}{4n^2}} \leq w$$

where n is the required sample size, p is the expected proportion, z is the critical value of the standard normal distribution, and w is the desired width of the CI.

Statistical analysis

Baseline characteristics of the RCTs were summarised descriptively by number (percentage) or median (IQR) based on the data type. Data extraction error rates were calculated at the level of data extraction cells in Excel sheets (for example, in an adverse event information extraction task for one RCT, if the total adverse event information contained 10 cells, with extraction errors in two cells regarding adverse event frequency, the error rate would be 20.0% (2/10)). Given the potential for clustering effects due to the diverse expertise and experience among the three people who verified Claude 3.5's extraction errors, a generalised linear mixed model with a logit link was used to estimate the error rates of data extractions and their 95% CIs. We summarised various error mechanisms and categorised them based on empirical observations. Then, we calculated the percentage of each error type relative to the total errors, as well as the distribution of different error types across various data elements.

Additionally, we also calculated recall and F1-scores for each data element to provide a more comprehensive performance evaluation; their definitions and detailed results are presented in online supplemental table 3. Data analysis was undertaken using Stata/SE V.15.0 (StataCorp) and R V.4.5.1 (R Foundation for Statistical Computing, Vienna, Austria). A two-sided $p<0.05$ was considered statistically significant.

Post hoc analysis

In order to explore whether trial reporting quality influenced LLM data extraction accuracy, we conducted a post hoc analysis, based on a subsample of 66 trials (10% of 664) that were selected by simple random sampling. Through team consensus, we identified 25 items (online supplemental table 4) most relevant to our data extraction framework, encompassing the Methods, Results and Open Science sections that stem from the updated Consolidated Standards of Reporting Trials (CONSORT) 2025 guideline.[30] Items were scored as follows: full adherence was assigned 1 point, partial adherence 0.5 points, and non-adherence 0 points. Items deemed non-applicable to specific trials were marked 'NA' and excluded from calculations. Scoring was performed independently by two reviewers (ZY and JHF), with an overall agreement of 91.5%. All discrepancies were resolved through discussion until consensus was reached. We assessed their adherence to the CONSORT 2025[30] reporting guidelines and used a prespecified threshold of 50% compliance to dichotomise trials into high-quality and low-quality reporting groups for exploring differences in extraction errors between groups. Risk differences (RD) with 95% CIs were calculated to compare error rates between the two groups.

Results

A total of 664 RCTs were included in the current study. As shown in online supplemental table 5, the median sample size was 72 (IQR: 40–134). These trials were predominantly conducted in Asia (19.9%, 132/664), the Americas (15.8%, 105/664) and Europe (10.7%, 71/664), although a substantial proportion (42.0%, 279/664) did not report their region of trial. Among these trials, 282 (42.5%) were single-centre trials, while 194 (29.2%) were multicentre trials. With regard to the design, most (89.5%, 594/664) were parallel group trials, followed by cross-over trials (9.6%, 64/664), then cluster trials (0.9%, 6/664). Additionally, only 27.1% (180/664) of trials reported registration status in their full texts.

Error rate of data extraction

We evaluated the error of data extraction performed by Claude 3.5 Sonnet across 23 069 data extraction cells from 664 RCTs. Overall, 1542 extraction errors were identified, yielding a total error rate of 6.6% (95% CI 5.4% to 8.2%) as shown in table 2. For adverse events, which constituted the majority of extracted data cells (12 445/23 069), the error rate was 7.6% (95% CI 4.9% to 11.8%).

Error rates varied substantially across different data elements. For the baseline characteristics, the highest error rates were observed for region (39.0%, 95% CI 34.7% to 43.4%) and centre information (27.3%, 95% CI 24.0% to 30.8%). Moderate error rates were found for funding information (7.1%, 95% CI 5.4% to 9.3%). Furthermore, the error rates for extracting information on the year of publication, journal names, registration details, registration numbers, study designs, intervention names, control names, sample sizes, mean population ages, and age ranges were all below 5.0%. Notably, no extraction errors were identified for trial titles, disease information, or data sharing statements.

Types of error

Based on the mechanisms underlying data extraction errors, we empirically classified these errors into five distinct categories: false absence reporting, misallocation, fabricated data, missed or omitted data and numerical error. Among these, misallocation was the most common, comprising 57.1% (881/1542) errors recorded in the studies. The second most frequent error was missed or omitted data, accounting for 23.2% (357/1542). Numerical errors, false absence reporting and fabricated data represented 15.2% (233/1542), 4.3% (28/1542) and 0.2% (3/1542), respectively (figure 1).

The distribution of extraction error types varied substantially across different data extraction elements (figure 2 and online supplemental Table 6). Misallocation was the most prevalent type of error; this error type was particularly pronounced in errors of region (87.6%, 227/259) and centre information (96.1%, 174/181). Missed or omitted data were observed exclusively in the extraction of adverse events and intervention/control drug names.

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Table 2 Performance of Claude 3.5 Sonnet in different data element extraction tasks

Domains of data extraction	No. of incorrect extractions	No. of total extractions	Error rate (95% CI)
Study baseline			
Title	0	664	--
Year	5	664	0.6% (0.1% to 2.7%)
Journal	1	664	0.2% (0.0% to 1.1%)
Region	259	664	39.0% (34.7% to 43.4%)
Centre	181	664	27.3% (24.0% to 30.8%)
Funding	47	664	7.1% (5.4% to 9.3%)
Trial registration	6	664	0.9% (0.4% to 2.0%)
Registration number	2	664	0.3% (0.1% to 1.2%)
Data sharing statement	0	664	--
Disease	0	664	--
Sample size	15	664	2.3% (1.4% to 3.7%)
Age–mean	12	664	1.8% (0.9% to 3.4%)
Age–range	25	664	3.8% (2.5% to 5.6%)
Intervention	12	664	1.8% (1.0% to 3.2%)
Control	9	664	1.4% (0.7% to 2.6%)
Study design	26	664	3.9% (2.5% to 6.0%)
Adverse event			
Adverse events	942	12 445	7.6% (4.9% to 11.8%)
Total	1542	23 069	6.6% (5.4% to 8.2%)

Importantly, 96.1% (343/357) of all data omissions occurred specifically during the extraction of adverse events information. Similarly, errors in adverse event frequency (80.0%, 188/235) data constituted a substantial proportion of all numerical errors.

Post hoc analysis: impact of reporting quality on error rate of data extraction

We conducted exploratory post hoc analyses to investigate whether trial reporting quality influenced LLM's data extraction error. The subsample for post hoc analysis comprised 66 trials published between 1961 and 2023. Nearly half (n=31, 47.0%) were published before 2010. Baseline characteristics are presented in online supplemental table 5. The CONSORT percentage complete ranged from 11.1% to 79.2%, with only 14 trials (21.2%) achieving scores of 50% or above. Stratified analysis revealed that Claude 3.5 Sonnet demonstrated significantly lower error rates in trials with high-quality reporting group compared with low-quality reporting group, for both overall extraction (RD=−0.04, 95% CI −0.06 to −0.02) and adverse events (RD=−0.06, 95% CI −0.09 to −0.04), but not for basic information extraction (RD=−0.02, 95% CI −0.05 to 0.01) (see figure 3).

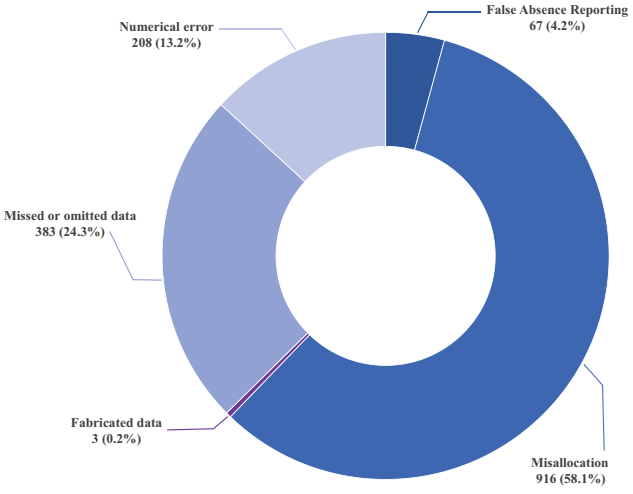


Figure 1 Proportion of 1542 data extraction error classified by five types of data extraction error.

Discussion

In this study, we evaluated the performance of Claude 3.5 Sonnet in the automatic data extraction from RCTs involving adverse drug events, with data extraction error as the primary consideration. The overall error rate for data extraction was 6.6%. Specifically, the error rate for extracting basic information was 5.7%, while for extracting adverse events it was 7.6%. Misallocation and missed or omitted data were identified as the two most prevalent types, accounting for 80.3% of the total errors, while fabricated data (0.2%) were rare. We found that the distribution of these error types varies across different data elements. Before using LLM for data extraction, understanding the potential error patterns and mechanisms associated with each type of data element is critical to improving extraction accuracy. Our findings also indicate that reporting quality may affect the accuracy of LLM-assisted data extraction. Trials with more standardised and complete reporting according to CONSORT guidelines appeared to facilitate more accurate extraction.

Our results showed Claude 3.5 Sonnet also demonstrated relatively good performance in large-scale data extraction validation compared with previous small-sample conceptual studies,[5 6 31] where the error rates for extracting basic information and adverse events were lower than 10%. A previous trial showed that, at the cell level, single human extraction had an error rate of 36.3%, and even with double extraction and adjudication, the error rate remained at 24.1%; in our study, Claude 3.5 Sonnet's single extraction achieved significantly lower error rates than both benchmarks.[3] In spite of the low error rates, we still noticed that there were relatively high data extraction errors in some specific data elements (such as region and centre information, with error rates of 39.0% and 27.3%, respectively). These findings suggest that systematic review teams should conduct pilot testing before

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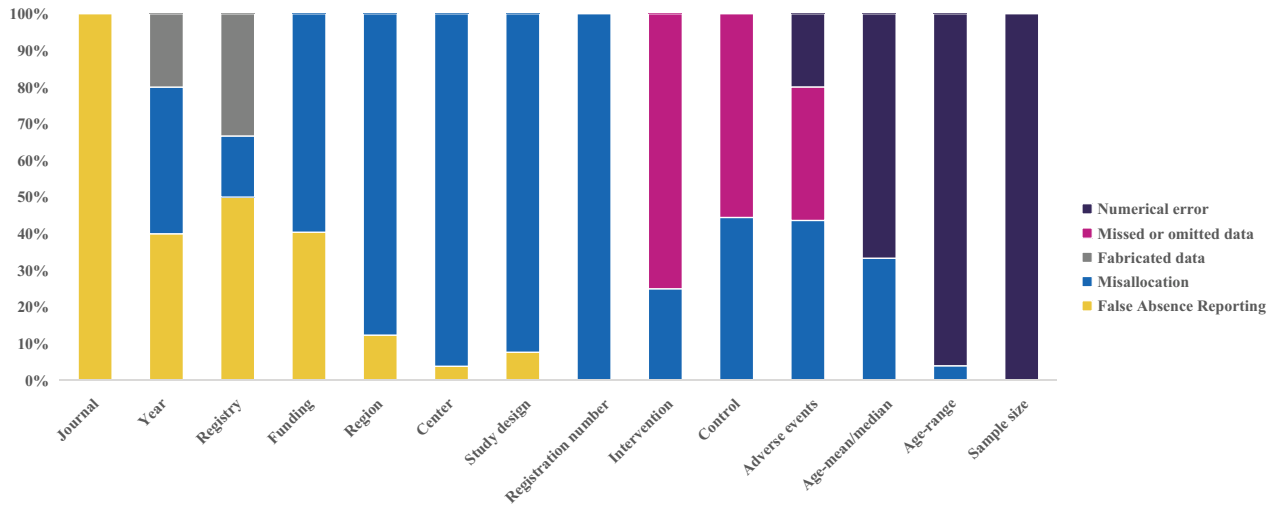


Figure 2 Percentage distribution of five types of data extraction error across 14 data elements with extraction errors.

formal extraction to identify LLM performance across different data elements, and develop tailored data validation strategies that focus more resources on data elements known to have higher error rates when using LLMs.

We empirically classified the LLM-based data extraction errors into five types based on their underlying mechanisms. Based on the classification, we further found that the prevalence of misallocation (57.1%) highlighted a fundamental challenge in Claude for data extraction: the difficulty of distinguishing between similar types of information presented in different contexts. For instance, despite providing detailed explanations and strict constraints in our prompts for extracting region (87.6%) and centre information (96.1%), Claude 3.5 Sonnet still confused authors' affiliated institutions with trial locations, resulting in higher identification errors. This suggests that general LLMs, such as Claude, may rely heavily on text recognition rather than true semantic understanding when identifying information that requires contextual interpretation, indicating that caution should be exercised when using crude LLMs for similar information extraction in subsequent tasks.

Furthermore, approximately one-fifth of errors involved missed or omitted data, particularly in adverse event extraction. These omissions likely stemmed from the inconsistent terminology and varied formats used to report adverse events in clinical trials.[32] More broadly, poor reporting quality appeared to exacerbate extraction challenges. Our post hoc analysis findings supported this hypothesis, demonstrating that Claude 3.5 Sonnet was more prone to data extraction errors in trials with poor reporting quality. Complete reporting of clinical trials requires more than journal articles alone,[33] and if trial data were standardised, machine-readable, and routinely shared alongside publications (eg, as structured spreadsheets), the need for complex extraction processes would be significantly reduced. The Transparency and Openness Promotion Guidelines provide a framework for advancing such practices, emphasising data, materials, and code sharing to enhance research transparency.[34]

Implications for future practice

The current study showed that LLM-assisted extraction with human verification achieved comparable accuracy to manual extraction with verification, while demonstrating superior efficiency.[35] Given these encouraging findings, further research is worthwhile to investigate optimal LLM-based extraction methods[36] (eg, different LLMs for parallel extraction, or using one LLM for extraction with another LLM adjudicating results, etc) while our findings provide practical guidance for evaluating and validating whether these

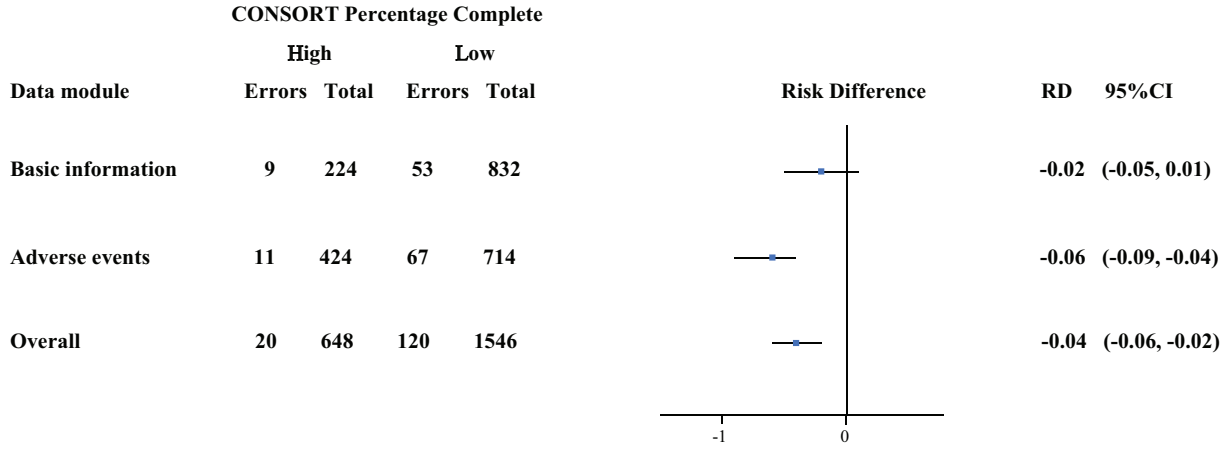


Figure 3 Comparison of Claude 3.5 extraction error rates in trials with high versus low CONSORT reporting quality. CONSORT, Consolidated Standards of Reporting Trials; RD, risk difference.

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methods could achieve dual-human extraction accuracy while improving efficiency. In our previous proof-of-concept study,[37] well-designed multiple-sentence prompts could provide more accurate data extraction for systematic reviews, while iterative prompting strategies may potentially hinder data extraction errors. Our current study focused on error rates and mechanisms in data extraction from RCTs via Claude 3.5 Sonnet. In the future, in-depth studies comparing LLM-based and human extraction methods should extend beyond extraction error to examine how these extraction errors affect evidence synthesis, such as meta-analyses, and ultimately influence systematic review conclusions.

Strengths and limitations

Our study is based on what we consider to be the largest databank of RCTs, with a sufficient number of randomly selected samples to ensure sufficient breadth and representativeness. The data collection process was closely supervised, including at least two rounds of checking to reduce human errors. Our study provides a detailed classification of error types and their distribution across various data elements in the field of adverse drug events for children, offering granular insights into the specific challenges of LLM-based data extraction from RCTs. These have significant implications for the integration of LLMs such as Claude into evidence synthesis workflows.

However, several limitations of this study also warrant consideration. First, the database used for analysing data extraction accuracy was established on 14 February 2023, and the included RCTs were restricted to drug-related trials, which may limit the generalisability of our results. Second, due to the inherent variability in outputs generated by LLM,[8] this study did not evaluate the consistency of responses to the same prompt across multiple iterations. However, we conducted repeated extraction tests on 10 RCTs from the pilot study, which demonstrated a high test–retest reliability (replication 1: 1.9%; replication 2: 1.3% error rates; online supplemental table 7). Third, we tested the data extraction performance of both Claude 3.5 Sonnet and ChatGPT-4 in our pilot study, selecting only Claude for the main study based on its superior performance while also considering time and cost factors. Fourth, our findings only reflected the performance level of LLMs available at the time of the study. Given the rapid advancement of artificial intelligence technologies, updated algorithms and evolving models will need to be continually evaluated in future research. Finally, we only tested Claude's performance on data extraction from main PDFs. As adverse events are often reported in appendices or supplementary materials, this may limit the representativeness of our findings, particularly for AE extraction in real-world systematic reviews where all available sources must be reviewed.

Conclusions

The empirical evidence from this study highlights the potential of Claude in automating data extraction for evidence synthesis. However, researchers should fully recognise its potential limitations before widespread adoption. Pilot testing is necessary to evaluate performance and optimise prompts prior to formal implementation.

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Contributors SF: conceptualisation, data analysis and result interpretation, manuscript drafting, data collection and manuscript editing. MC: data collection, data analysis and result interpretation, manuscript editing. SAD: statistical guidance, methodology guidance, manuscript editing. ZY: data collection, manuscript editing. ZP and YT: methodology guidance, manuscript editing. CZ, LF-K, LL, EM-W and MHM: statistical guidance, methodology guidance, manuscript editing. XM and CX: methodology guidance, result interpretation, manuscript editing. All authors read and approved the final manuscript. SF is the guarantor. This is a study on AI. We used Claude 3.5 Sonnet to extract information as part of the project. We did not use AI for purposes other than the research project. We employed Claude 3.5 Sonnet to perform data extraction tasks, aiming to investigate error patterns of large language models in extracting data from randomised controlled trials in the medical domain.

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References

1 **Xu C**, Yu T, Furuya-Kanamori L, *et al*. Validity of data extraction in evidence synthesis practice of adverse events: reproducibility study. *BMJ* 2022;377:e069155.
2 **Li T**, Higgins JPT, Deeks JJ. Chapter 5: collecting data. In: *Cochrane Handbook for Systematic Reviews of Interventions Version 6.5*. Cochrane, 2019. Available: https://training.cochrane.org/handbook/current/chapter-05
3 **Tang L**, Wang R, Doi SAR, *et al*. Effects of double data extraction on errors in evidence synthesis: a crossover, multicenter, investigator-blinded, randomized controlled trial. *Postgrad Med J* 2025;101:603–11.
4 **Lin Z**. How to write effective prompts for large language models. *Nat Hum Behav* 2024;8:611–5.
5 **Gartlehner G**, Kahwati L, Hilscher R, *et al*. Data extraction for evidence synthesis using a large language model: A proof-of-concept study. *Res Synth Methods* 2024;15:576–89.
6 **Konet A**, Thomas I, Gartlehner G, *et al*. Performance of two large language models for data extraction in evidence synthesis. *Res Synth Methods* 2024;15:818–24.
7 **Mahmoudi H**, Chang D, Lee H, *et al*. A critical assessment of large language models for systematic reviews: utilizing ChatGPT for complex data extraction. *SSRN* 2024.
8 **Gartlehner G**, Kahwati L, Nussbaumer-Streit B, *et al*. From promise to practice: challenges and pitfalls in the evaluation of large language models for data extraction in evidence synthesis. *BMJ Evid Based Med* 2025;30:385–9.
9 **Gartlehner G**, Kugley S, Crotty K, *et al*. Artificial Intelligence-Assisted Data Extraction With a Large Language Model: A Study Within Reviews. *Ann Intern Med* 2025;178:1763–71.
10 **Xu C**, Doi SA, Vohra S, *et al*. The Trial Bank initiative to promote trustworthy living evidence for medication safety (PROMISE) of children: design and pilot validation. *BMC Med* 2026;24:181.
11 **Sounderajah V**, Guni A, Liu X, *et al*. The STARD-AI reporting guideline for diagnostic accuracy studies using artificial intelligence. *Nat Med* 2025;31:3283–9.
12 **Fan S**. Data from: evaluating data extraction error by a large language model from randomized controlled trials. Open Science Framework; 2026.
13 **Ábarzúa-Araya A**, Navarrete-Dechent CP, Heusser F, *et al*. Atenolol versus propranolol for the treatment of infantile hemangiomas: a randomized controlled study. *J Am Acad Dermatol* 2014;70:1045–9.
14 **Abdelazim R**, Salah D, Labib HA, *et al*. Methylene blue compared to norepinephrine in the management of vasoplegic syndrome in pediatric patients after cardiopulmonary bypass: a randomized controlled study. *Egyptian Journal of Anaesthesia* 2016;32:269–75.
15 **Abdelmawgoud A**, Mohy A. Effect of oral dextromethorphan versus oral ketamine on sevoflurane related emergence agitation in children undergoing adenotonsillectomy. *Egyptian Journal of Anaesthesia* 2012;28:243–8.
16 **Al-Layla A**, Mahafza TM. Antibiotics do not reduce post-tonsillectomy morbidity in children. *Eur Arch Otorhinolaryngol* 2013;270:367–70.
17 **Abeyagunawardena AS**, Trompeter RS. Increasing the dose of prednisolone during viral infections reduces the risk of relapse in nephrotic syndrome: a randomised controlled trial. *Arch Dis Child* 2008;93:226–8.
18 **Abo-EL-Roos MA**, El-Negmy EH, El-Shemy SA, *et al*. Assessment of Interleukin-6 in Young Swimmers Suffering from Myofascial Pain Syndrome Using Lidocaine Phonophoresis: A Randomized Controlled Trial. *Int J Res Pharm Sci* 2020;11:2731–40.
19 **Adegnika AA**, Zinsou JF, Issifou S, *et al*. Randomized, controlled, assessor-blind clinical trial to assess the efficacy of single- versus repeated-dose albendazole to treat ascaris lumbricoides, trichuris trichiura, and hookworm infection. *Antimicrob Agents Chemother* 2014;58:2535–40.
20 **Agarwal V**, Sitholey P, Kumar S, *et al*. Double-blind, placebo-controlled trial of clonidine in hyperactive children with mental retardation. *Ment Retard* 2001;39:259–67.
21 **Agweyu A**, Gathara D, Oliwa J, *et al*. Oral amoxicillin versus benzyl penicillin for severe pneumonia among Kenyan children: a pragmatic randomized controlled noninferiority trial. *Clin Infect Dis* 2015;60:1216–24.
22 **Ahmadiafshar A**, Mogimi Hadji M, Rezaei N. Comparison of Effectiveness between Beclomethasone Dipropionate and Fluticasone Propionate in Treatment of Children with Moderate Asthma. *World Allergy Organ J* 2010;3:250–2.
23 **Zorzela L**, Loke YK, Ioannidis JP, *et al*. PRISMA harms checklist: improving harms reporting in systematic reviews. *BMJ* 2016;352:i157.
24 **Cheng D**, Liu L, Hu Z. Prevention of anesthesia-induced injection pain of propofol in pediatric anesthesia. *Pak J Med Sci* 2017;33:752–6.
25 **Gregorio GV**, Jones H, Choudhuri K, *et al*. Autoantibody prevalence in chronic hepatitis B virus infection: effect in interferon alfa. *Hepatology* 1996;24:520–3.
26 **Chawes BLK**, Piccinno A, Kreiner-Møller E, *et al*. Pharmacokinetic comparison of inhaled fixed combination vs. the free combination of beclomethasone and formoterol pMDIs in asthmatic children. *Br J Clin Pharmacol* 2013;75:1081–8.
27 **Ergil J**, Akkaya T, Gozaydin O, *et al*. Vasoconstrictive and analgesic efficacy of locally infiltrated levobupivacaine in tonsillectomy patients. *Int J Pediatr Otorhinolaryngol* 2012;76:1429–33.
28 **Jørgensen F**, Andersson B, Hanson LA, *et al*. Gamma-globulin treatment of recurrent acute otitis media in children. *Pediatr Infect Dis J* 1990;9:389–94.
29 **Brown LD**, Cai TT, DasGupta A. Interval Estimation for a Binomial Proportion. *Statist Sci* 2001;16:101–33.
30 **Hopewell S**, Chan A-W, Collins GS, *et al*. CONSORT 2025 statement: updated guideline for reporting randomised trials. *BMJ* 2025;389:e081123.
31 **Kim JK**, Chua ME, Li TG, *et al*. Novel AI applications in systematic review: GPT-4 assisted data extraction, analysis, review of bias. *BMJ Evid Based Med* 2025;30:313–22.
32 **Phillips R**, Hazell L, Sauzet O, *et al*. Analysis and reporting of adverse events in randomised controlled trials: a review. *BMJ Open* 2019;9:e024537.
33 **Mayo-Wilson E**. Complete reporting of clinical trials requires more than journal articles. *BMJ* 2025;389:r494.
34 **Grant S**, Corker KS, Mellor D, *et al*. TOP 2025: an update to the transparency and openness promotion guidelines. 2025. Available: https://osf.io/preprints/metaarxiv/nmfs6_v3
35 **Helms Andersen T**, Marcussen TM, Termannsen AD, *et al*. Using Artificial Intelligence Tools as Second Reviewers for Data Extraction in Systematic Reviews: A Performance Comparison of Two AI Tools Against Human Reviewers. *Cochrane Evidence Synthesis and Methods* 2025;3:e70036.
36 **Peng Z**, Fan S, Tian Y, *et al*. Comparing the accuracy of AI-assisted data extraction versus human double extraction in evidence synthesis: a randomised controlled trial protocol. *BMJ Open* 2025;15.
37 **Peng Z**, Wu X, Qin Z, *et al*. Accuracy of large language models in data extraction from randomized controlled trials in sleep medicine: A proof-of-concept study. *Sleep Med Rev* 2025;84:102192.

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